



Assignments

Pointers



Q1. Write a program to find out the product of two numbers using pointers.

Solution:

```
#include <bits/stdc++.h>
using namespace std
int main() {
    int x, y;
    cin >> x >> y;
    int *ptrX, *ptrY;
    ptrX = &x;
    ptrY = &y;
    int ans;
    int *ptrAns = &ans;
    *ptrAns = (*ptrX) * (*ptrY);
    cout << *ptrAns << endl;
    return 0;
}
```

Q2. int *p, q; What does both of these terms signify?

Solution:

p is a pointer and q is an integer. * is grouped with variables, not data types.

Q3. Predict the output of the following code snippet.

```
int a = 10, b = 20;
int *ptr = &a;
b = *ptr + 1;
ptr = &b;
cout << *ptr << ' ' << a << ' ' << b;
```

Solution:

11 10 11

ptr points to a, b = a + 1, so ptr points to b

Q4. Predict the output of the following code snippet.

```
int a = 15, b = 20;
int *ptr = &a;
int *ptr2 = &b;
*ptr = *ptr2;
```

Solution:

a gets the value of b

Q5. Predict the output of the following code snippet.

```
int a = 10, b = 20;  
int *ptr;  
*ptr = 5;
```

Solution:

The program is correct and will compile but might result in runtime error. This is because ptr will have a garbage address which might even point to a location which does not belong to our program and hence might result in Segmentation Fault when we are trying to access it.





**THANK
YOU !**

